

LAB | Port Mapping

```
seife@SeifEddine:~$ docker run -dit --name web-nomap --rm ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
51fe6271825457c63677e3d13f8342b7a3e2f6a061ac37ea71838b5f50dd24d4
seife@SeifEddine:~$ docker exec web-nomap apt-get update
docker exec web-nomap apt-get install -y nginx curl
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
```

```
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for ca-certificates (20260223) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
seife@SeifEddine:~$
```

```
seife@SeifEddine:~$ docker exec -d web-nomap nginx -g daemon o
ff
seife@SeifEddine:~$ docker exec web-nomap bash -lc 'echo "<h1>
Hello from INSIDE the container (no mapping)</h1>" > /var/www/
html/index.html'
seife@SeifEddine:~$ docker exec web-nomap bash -lc 'echo "<h1>
Hello from INSIDE the container (no mapping)</h1>" > /var/www/
html/index.html
> docker exec -d web-nomap nginx -g daemon off
> docker exec web-nomap bash -lc 'echo "<h1>Hello from INSIDE
> docker exec web-nomap bash -lc 'echo "<h1>Hello from INSIDE
the container (no mapping)</h1>" > /var/www/html/index.html'^C
seife@SeifEddine:~$ docker exec web-nomap bash -lc 'echo "<h1>
Hello from INSIDE the container (no mapping)</h1>" > /var/www/
html/index.html'
seife@SeifEddine:~$ docker exec -d web-nomap nginx -g "daemon
off;"
seife@SeifEddine:~$ <h1>Hello from INSIDE the container (no ma
pping)</h1>
-bash: syntax error near unexpected token `('
seife@SeifEddine:~$ docker exec web-nomap curl -s http://local
host | head -n1
<h1>Hello from INSIDE the container (no mapping)</h1>
seife@SeifEddine:~$
```

```
seife@SeifEddine:~$ curl -v http://localhost
* Host localhost:80 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:80...
* Connected to localhost (::1) port 80
> GET / HTTP/1.1
> Host: localhost
> User-Agent: curl/8.5.0
> Accept: */*
>
< HTTP/1.1 200 OK
< Date: Sat, 16 May 2026 14:52:44 GMT
< Server: Apache/2.4.58 (Ubuntu)
< Last-Modified: Thu, 30 Apr 2026 17:50:34 GMT
< ETag: "32-650b119f06294"
< Accept-Ranges: bytes
< Content-Length: 50
< Content-Type: text/html
<
* Connection #0 to host localhost left intact
<h1>Hello from Ansible-managed Apache server!</h1>seife@SeifEd
dine:~$
```

 Sign In

```
* Connection #0 to host localhost left intact
<h1>Hello from Ansible-managed Apache server!</h1>seife@SeifEd
dine:~$ docker rm -f web-nomap
web-nomap
seife@SeifEddine:~$ docker run -dit --name web-mapped --rm -p
8080:80 ubuntu
0f943b473654c3b3ce004f4f398e17a0a2807926662179377da4545756bb0f
06
seife@SeifEddine:~$ docker exec web-mapped apt-get update
docker exec web-mapped apt-get install -y nginx curl
Get:1 http://archive.ubuntu.com/ubuntu InRelease [136
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Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
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done.
seife@SeifEddine:~$ docker exec web-mapped bash -lc 'echo "<h1
>Hello from a mapped port (8080->80)</h1>" > /var/www/html/inde
x.html'
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seife@SeifEddine:~$ docker exec web-mapped bash -lc 'echo "<h1>Hello from a mapped port (8080->80)</h1>" > /var/www/html/index.html'
seife@SeifEddine:~$ docker exec -d web-mapped nginx -g "daemon off;"
seife@SeifEddine:~$ docker exec web-mapped curl -s http://localhost | head -n1
<h1>Hello from a mapped port (8080->80)</h1>
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seife@SeifEddine:~$ docker exec web-mapped bash -lc 'echo "<h1>Hello from a mapped port (8080->80)</h1>" > /var/www/html/index.html'
seife@SeifEddine:~$ docker exec -d web-mapped nginx -g "daemon off;"
seife@SeifEddine:~$ docker exec web-mapped curl -s http://localhost | head -n1
<h1>Hello from a mapped port (8080->80)</h1>
seife@SeifEddine:~$ curl -s http://localhost:8080 | head -n1
<h1>Hello from a mapped port (8080->80)</h1>
seife@SeifEddine:~$ docker ps --filter name=web-mapped
# or
docker inspect -f '{{json .NetworkSettings.Ports}}' web-mapped | jq
CONTAINER ID    IMAGE    COMMAND                  CREATED        STATUS
PORTS          NAMES
0f943b473654    ubuntu  "/bin/bash"             3 minutes ago Up 3 minutes
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp web-mapped
Command 'jq' not found, but can be installed with:
sudo snap install jq # version 1.5+dfsg-1, or
sudo apt install jq # version 1.7.1-3ubuntu0.24.04.2
See 'snap info jq' for additional versions.
seife@SeifEddine:~$
```

```
minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp web-mapped
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See 'snap info jq' for additional versions.
seife@SeifEddine:~$ docker rm -f web-mapped
web-mapped
seife@SeifEddine:~$
```